

OvidAI

PRODUCT & INFRASTRUCTURE BLUEPRINT

CONFIDENTIAL · STRATEGY DOCUMENT · V1.0

Zero to 10 Million Users

The complete execution plan to transform OvidAI from a pixel-perfect UI prototype into a production-grade, AI-native super-app — powered almost entirely by **Google's free-tier infrastructure**, a freemium **Ovid Provider** model gateway, and a privacy-first sign-in that syncs only what users love: their chat history.

10M

TARGET USERS

\$0 →

INFRA COST AT
LAUNCH

15+

AI PROVIDERS
(BYOK)

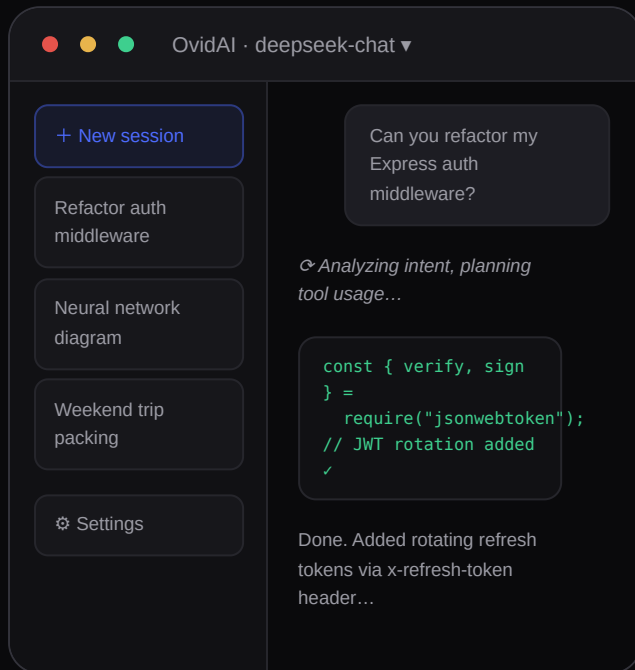
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EXECUTION
PHASES

A Beautiful Shell, Waiting for a Brain

OvidAI today is a complete, premium UI/UX prototype (Flutter, ~3,000 lines) with every screen designed — but all data is in-memory demo content. That is our advantage: zero legacy debt, clean slate architecture.

THE APP TODAY — CHAT-FIRST SHELL



Design system is production-ready. "Aether" — monochrome dark, single indigo accent (#4D6BFE), Inter + JetBrains Mono, DeepSeek-grade polish. No redesign needed to scale.

SIX SCREENS ALREADY DESIGNED



Chat

Sessions, model picker, reasoning chips, code blocks, image-gen cards, attach & voice.



Studio

GitHub-bound file tree, editor tabs, sandbox terminal with agent output.



Browser

Manus-style agent browser — AI logs into sites, user can take over.



Plugins

Marketplace: OpenCode agent, MCP hub, sandbox runtime, 30+ community plugins.



Providers

15 BYOK providers — OpenAI, Anthropic, Gemini, DeepSeek, Groq, Ollama...



Usage

Per-client token & cost analytics across apps and models.

WHAT'S MISSING (THE "BRAIN")

- ▶ **No backend** — all state is in-memory, lost on restart
- ▶ **No auth** — no user identity, no sync
- ▶ **No live inference** — replies are hardcoded demo strings
- ▶ **No free models** — users must bring their own keys (friction!)
- ▶ **No persistence** — chat history dies with the app

Free AI for Everyone. Keys for Power Users.

The single biggest growth unlock: a new user opens the app and gets a working AI reply in under 10 seconds — no API key, no credit card. We become the "Ovid Provider" — our own free model gateway, exactly like OpenCode ships free models.

THE THREE-LAYER MODEL STRATEGY

 **Ovid Free** DEFAULT

Our own gateway proxies **genuinely free models** — Gemini 2.5 Flash (AI Studio free tier), Groq Llama, Cerebras, GitHub Models, OpenRouter free variants.

Per-user daily quota (e.g. 50 msgs/day) enforced by Firebase Auth UID. No key ever needed.

 **BYOK** POWER

The 15 providers already in the UI. User pastes their own key → **unlimited usage, zero cost to us**. Keys stay on-device (flutter_secure_storage).

This is our margin protector — heavy users self-fund.

 **Ovid Pro** ₹/\$ MRR

Phase 4+: pooled premium models (GPT-5, Claude, Gemini Pro) with generous limits, priority queue, image gen, longer memory.

\$5–8/mo via Play Billing — only pitched after the habit forms.

THE PRIVACY-FIRST AUTH PROMISE (OUR DIFFERENTIATOR)

✓ **What syncs to cloud — chat history only**

- ▶ **Chat sessions + messages** → Firestore, per-user, encrypted at rest
- ▶ **Session titles & model choice** → seamless multi-device resume
- ▶ **Free-tier quota counter** → per-UID daily allowance

✗ **What NEVER leaves the device**

- ▶ **API keys (BYOK)** — secure enclave / EncryptedSharedPreferences only
- ▶ **Code, files, repos** — Studio syncs via user's own GitHub, not us
- ▶ **Browser sessions/cookies** — on-device WebView only

The sign-in ask is tiny, the value is huge: "Sign in with Google to save your chats & unlock free AI models." One tap (Firebase Auth Google Sign-In), no password, no card. Code history is explicitly excluded — a trust signal developers will love and share.

WHY THIS WINS AGAINST THE ALTERNATIVES

COMPETITOR PATTERN	THEIR FRICTION	OVIDAI ANSWER
ChatGPT / Claude apps	Single model, paywall for best models, closed	15+ providers, free gateway, open plugins
Ovid AI: Yes, no, how? Blueprint	Dead on arrival — no key = no value	Works instantly with Ovid Free models
OpenCode / OpenAI	Desktop only, developer only	Mobile first, chat first, for everyone

100% Google Free-Tier Infrastructure

Every layer maps to a Google product with a genuinely usable free tier. At launch we pay \$0. The architecture scales to 10M users on the same primitives — we only start paying when revenue already exists.

◆ CLIENT LAYER — FLUTTER APP (WHAT USERS TOUCH)

Chat UI

Existing screens, streaming via SSE/WebSocket

Secure Vault

BYOK keys on-device only (Keystore)

Offline Cache

Hive/Isar mirror of Firestore chats

Studio + Browser

Sandbox, agent WebView — client-side



◆ IDENTITY & SYNC — FIREBASE (SPARK FREE PLAN)

Firebase Auth

Google Sign-In, 1 tap. Free: unlimited MAU

Cloud Firestore

Chat history per UID. Free: 1GB + 50k reads/day

FCM Push

Agent-done alerts, re-engagement. Free: unlimited

Remote Config

Model catalog, quotas, kill-switches. Free



◆ OVID PROVIDER GATEWAY — CLOUD RUN (SERVERLESS)

Cloud Run API

Free: 2M req/mo. OpenAI-compatible /v1/chat

Quota Guard

Verify Firebase ID token → check daily limit

Model Router

Gemini Flash → Groq → Cerebras failover chain

Key Pool

Rotating free-tier keys in Secret Manager



◆ INTELLIGENCE — FREE MODEL SUPPLY (UPSTREAM)

Gemini 2.5 Flash

AI Studio free tier — primary workhorse

Groq LPU

Llama 3.3 70B free tier — speed demon

Cerebras / GitHub

Free-tier overflow capacity

Gemini Pro (Pro tier)

Paid upsell path, Vertex AI later

\$0 at launch.

Auth unlimited, FCM unlimited, Firestore 1GB, Cloud Run 2M req — all inside free tiers for first ~50k users.

Scale = dial, not rebuild.

Same services, Blaze pay-as-you-go kicks in gradually. Firestore + Cloud Run handle 10M users by design.

Google for Startups.

Apply for Cloud credits (\$2k–\$100k tier) to make even the paid phase effectively free for year one.

Our Free Model Gateway, Explained

Exactly how OpenCode ships free models: the app talks to one OpenAI-compatible endpoint — ours. The gateway hides every upstream provider, enforces fair limits, and fails over invisibly. Users just see "Ovid Free • online".

ONE REQUEST'S JOURNEY (UNDER 2 SECONDS)



FREE QUOTA DESIGN (PER USER / DAY)

MODEL TIER	EXAMPLES	FREE LIMIT
Fast	Gemini 2.5 Flash, Llama 3.3 (Groq)	50 msgs
Smart	DeepSeek-R1 (GitHub), gpt-oss-120b	15 msgs
Image	Pollinations / HF Spaces	5 gens
BYOK	Any of 15 providers	Unlimited

Limits live in Remote Config — tune daily without shipping an app update. Generous on launch day, sustainable forever.

WHY UPSTREAM STAYS FREE FOR US

- ▶ **Gemini AI Studio free tier** — generous RPD per key; pool N keys behind the gateway, rotate on 429
- ▶ **Groq / Cerebras free tiers** — high-RPS LPU inference costs them little, costs us nothing
- ▶ **GitHub Models** — free with GitHub token; OAuth'd users bring quota with them
- ▶ **OpenRouter :free variants** — DeepSeek/Llama free endpoints as final fallback
- ▶ **Cache aggressively** — identical FAQ-style prompts served from Firestore cache, zero inference cost

Cost math at 1M MAU: ~92% of requests hit free upstreams or cache → blended inference cost ≈ **< \$0.001/user/mo**, covered by Pro subscriptions 100× over.

WHAT THE USER SEES (ALREADY FITS OUR MODEL PICKER UI)

FREE

Ovid Provider — appears at top of the existing model sheet:

ovid-flash (Gemini 2.5)

ovid-think (DeepSeek-R1)

ovid-turbo (Llama 3.3 70B)

ovid-canvas (image gen)

Phase 0 & 1 — Foundation to Launch

From today's UI-only prototype to a live Play Store app with real AI replies, in two focused sprints. Everything here runs on \$0 infrastructure.

PHASE

0

Weeks 1–4



Foundation — Make It Real

0 → 1K users

Wire the beautiful shell to a beating heart. Ship to closed beta (Play Internal Testing) and dogfood hard.

- ▶ **Firebase Auth** — 1-tap Google Sign-In gate (skip allowed, quota smaller)
- ▶ **Firestore sync** — sessions/messages per UID, offline cache with Hive
- ▶ **Cloud Run gateway v1** — /v1/chat/completions, Gemini Flash only
- ▶ **Streaming replies** — replace receiveDemoReply() with SSE client
- ▶ **Quota counter** — 50 msg/day, Remote Config driven
- ▶ **Crashlytics + Analytics** — free, from day one

✓ Real AI reply < 2s

✓ Chats survive reinstall

✓ Crash-free > 99%

✓ Infra cost \$0

PHASE

1

Months 2–3



Public Launch — The Free AI Moment

1K → 100K users

Play Store public release. The hook: "ChatGPT-style app, but free, open, and yours." Ovid Provider goes multi-upstream with failover.

- ▶ **Model Router v2** — Gemini → Groq → Cerebras → GitHub failover chain
- ▶ **ovid-think** — free reasoning model with visible thinking chips (UI already exists!)
- ▶ **Image gen live** — Pollinations free endpoint into existing image card
- ▶ **BYOK activation** — Providers screen fully functional, keys on-device
- ▶ **FCM re-engagement** — "Your agent finished" + daily idea prompts
- ▶ **Referral loop** — share a chat as a beautiful public link (Firestore + hosting)

✓ D1 retention > 40%

✓ K-factor > 0.4 via shared chats

✓ Play rating > 4.5

✓ Cost still ≈ \$0

LAUNCH GROWTH ENGINE (NO AD SPEND)



Shareable chats

Every shared answer links back to the app — each user is a distribution node.



Free is the marketing

"50 free AI messages a day, no card" is a headline that spreads itself on X/Reddit.



Dev credibility

BYOK + MCP + OpenCode plugin wins the HN/dev crowd who evangelize hardest.



ASO + locales

Hindi + English at launch; India-first pricing psychology for Pro later.

Phase 2–4 — Agents, Monetization, 10M Scale

Turn a chat app into an agent platform, flip on revenue only after the habit is locked, then ride the same Google primitives to eight figures of users.

PHASE

2

Months 4–6



Agent Platform — Beyond Chat

100K → 1M users

Activate the screens we already designed: Studio, Browser, Plugins become real. This is the moat — nobody else has mobile-native agents with a sandbox.

- ▶ **Sandbox Runtime live** — proot/Termux userland; agent runs python/node on-device, \$0 server cost
- ▶ **OpenCode agent plugin** — full coding agent against user's GitHub repo
- ▶ **Agent browser** — WebView automation with user take-over (UI ready)
- ▶ **MCP Hub** — connect filesystem, GitHub, Postgres servers from the Plugins screen
- ▶ **Plugin submissions** — community plugins via GitHub PR → Remote Config catalog
- ▶ **Web + desktop wrappers** — Flutter's single codebase, near-zero extra cost

✓ Agent tasks/user/wk > 3

✓ 3rd-party plugins > 50

✓ Studio MAU > 15%

PHASE

3

Months 7–9



Monetization — Ovid Pro

1M → 3M users

Revenue appears only after daily habit exists. Pro = premium pooled models + headroom, sold via Play Billing (Google handles payments, tax, UPI).

- ▶ **Ovid Pro \$5–8/mo** — Gemini Pro, GPT-5-class pool, 10× limits, priority queue
- ▶ **Smart paywall** — shown only when free quota exhausts (moment of maximum intent)
- ▶ **Vertex AI migration** — paid Gemini via Vertex for Pro pool, committed-use discounts
- ▶ **Team seats** — shared workspaces, pooled BYOK — \$12/seat B2B wedge

✓ Free → Pro > 3%

✓ MRR > infra cost × 5

✓ Churn < 4%/mo

PHASE

4

Months 10–18



Scale — 10 Million Users

3M → 10M users

Same architecture, bigger dials. Google Cloud's committed-use + startup credits keep unit economics positive the whole way.

- ▶ **Multi-region Cloud Run** — India, US, EU; <300ms gateway latency globally
- ▶ **Memory (RAG plugin)** — vector recall via Firestore vector search, per-user isolated
- ▶ **Ovid API for devs** — sell our gateway itself; every client app = new distribution
- ▶ **On-device models** — Gemini Nano via AICore for offline/free-tier overflow

✓ 10M installs

✓ Infra < \$0.01/user/mo

✓ Pro ARPU sustains free tier

Unit Economics That Actually Work

Free users cost us almost nothing because compute happens on-device, on free upstream tiers, or on the user's own keys. Revenue only needs to cover the thin gateway layer.

COST PER USER PER MONTH (AT SCALE)

Auth	\$0	Firebase, unlimited
Chat sync	~\$0.001	Firestore Blaze
Gateway	~\$0.002	Cloud Run requests
Inference	~\$0.004	92% free/cache hits
Push & CDN	\$0	FCM unlimited

Blended cost ≈ \$0.007 / free user / month. A 3% Pro conversion at \$6/mo yields ~\$0.18 per total user — a 25x margin over cost. The free tier is self-funding.

TOP RISKS & OUR MITIGATIONS

⚠ Upstream free tiers shrink

Multi-upstream router + Remote Config model catalog = swap providers with zero app update. On-device Gemini Nano as permanent fallback.

⚠ Quota abuse / bot farms

Firebase App Check (free) attests real devices; per-UID + per-IP limits; anonymous accounts get half quota, nudging sign-in.

⚠ Gateway becomes a cost center

Aggressive semantic cache, prompt-length caps on free tier, and BYOK upsell nudges for heavy users ("add your key → unlimited").

WHERE EACH GOOGLE TIER BREAKS (AND COSTS \$0 UNTIL THEN)

SERVICE	FREE CEILING	≈ USERS COVERED
Firebase Auth	Unlimited (Google provider)	∞
FCM	Unlimited messages	∞
Firestore	50k reads + 20k writes /day	~15–25k DAU
Cloud Run	2M requests /mo	~30–50k DAU
Remote Config	Unlimited	∞

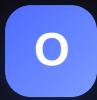
⚠ Trust: "you see my chats"

Public privacy contract: keys & code never leave device (verifiable — open-source the client). Chat sync is opt-out-able, deletable, exportable.

⚠ Play policy on AI apps

Content filters at gateway layer, report/block flows, clear AI-generated labeling — compliant from day one.

The one-line strategy: Google runs our backend for free, users' own devices run our agents for free, upstream labs subsidize our models for free — and we charge only the 3% who want the very best.



OvidAI

THE PATH FORWARD

"The UI is done. The design is done.

Now we give it a **heartbeat** — one tap of Google Sign-In, one free model, one delighted user at a time."

THIS WEEK

Create the Firebase project

Auth + Firestore + FCM in one console, one afternoon. The chat history promise starts here.

THIS MONTH

Ship the Ovid Gateway

One Cloud Run service, one free Gemini key, streaming replies into the existing chat UI.

THIS QUARTER

Launch on Play Store

"Free AI chat, 15 providers, your keys stay yours." Let the product be the marketing.

P0

FOUNDATION
· WK 1–4

P1

LAUNCH · MO
2–3

P2

AGENTS · MO
4–6

P3

PRO · MO 7–9

P4

10M · MO 10–
18